

Theory of production and cost

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INTRODUCTION

- Production is the process by which factor inputs are transformed into output. An increase in the quantity of factor inputs will lead to an increase in output. The theory of production is the study of how the output level changes as the quantity of factor inputs changes. To increase output, firms need to employ more factor inputs which will lead to an increase in costs. The theory of costs is the study of how the cost of production changes as the output level changes.

What is production?

- Production is an activity of utter importance for any economy. In fact, a nation with a high level of productive activities spearheads the prosperity charts. This is because raw goods, surely are valuable, but production done upon these raw goods adds up to their value or their want-satisfying power.
- *The production function can be denoted as*
- *$Q = f\{l, k\}$, where l is land and k is capital, the inputs of production*

Fixed and variable inputs

- A fixed input is one whose supply is inelastic in short run.
- A variable input is defined as one whose supply in the short run is elastic , e.g. labour, raw material etc.
- A fixed input remain fixed upto a certain level of output whereas a variable input changes with change in output.

Note

- **The short run** is the time period during which at least one of the factor inputs used in the production process is fixed. It does not correspond to any specific number of weeks, months or years as it varies from firm to firm and from industry to industry. For example, a web hosting firm may take only a few weeks or even days to increase its production capacity by purchasing more servers. However, an oil refining firm may take many years to increase its production capacity due to the long time period needed to build oil refineries.

THE SHORT-RUN THEORY OF PRODUCTION

- **Marginal Returns**
- If a firm wants to increase output, it can almost immediately employ more labour. However, it will not be able to employ more capital in the same time frame as acquisition of capital takes time. In economics, we distinguish between two types of factor inputs: variable factor input and fixed factor input. Variable factor inputs are factor inputs whose quantities can be changed in the short run. An example is labour. Fixed factor inputs are factor inputs whose quantities are fixed in the short run. An example is capital.

Example

Capital	Labour	Total Output	Additional Output
5	0	0	—
5	1	3	3
5	2	7	4
5	3	12	5
5	4	18	6
5	5	22	4
5	6	24	2
5	7	24	0
5	8	23	-1
5	9	20	-3

Explanation

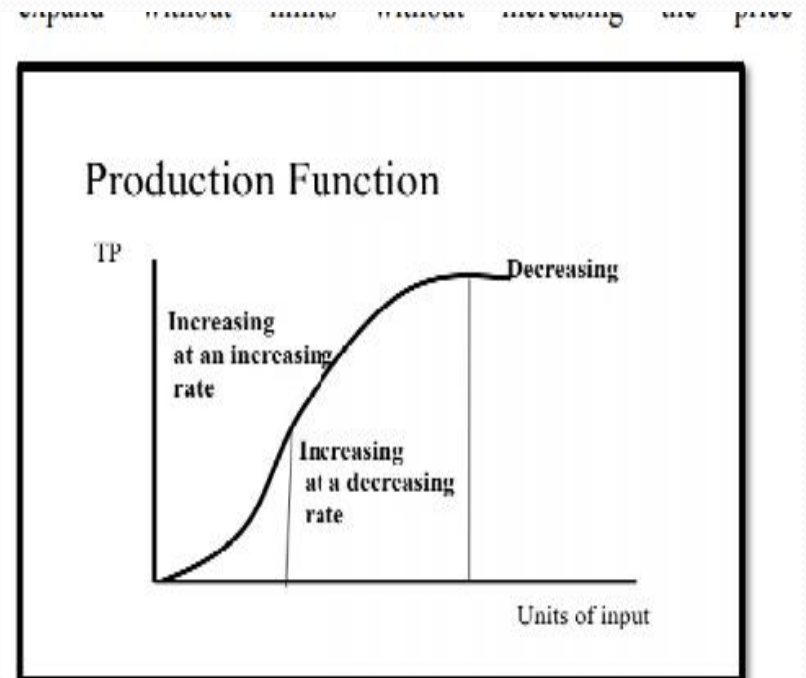
- In the above table, from the first unit of labour to the fourth, each additional unit of labour is adding more to total output than the previous additional unit and hence the firm is experiencing increasing marginal returns. Increasing marginal returns occur when each additional unit of a variable factor input (e.g. labour) is adding more to total output than the previous additional unit. This occurs due to under-utilisation of the fixed factor inputs (e.g. capital). However, from the fifth unit of labour onwards, each additional unit of labour is adding less to total output than the previous additional unit and hence the firm is experiencing diminishing marginal returns. Diminishing marginal returns occur when each additional unit of a variable factor input (e.g. labour) is adding less to total output than the previous additional unit. This occurs due to over-utilisation of the fixed factor inputs (e.g. capital). Diminishing marginal returns set in when the fifth unit of labour is employed. Furthermore, the seventh unit of labour is actually redundant. Total output even falls when the eighth unit of labour is employed.

The law of diminishing marginal returns

- The law of diminishing marginal returns states that if an increasing quantity of a variable factor input is used with a constant quantity of fixed factor inputs, an output level point will be reached beyond which each additional unit of the variable factor input will add less to total output than the previous additional unit.

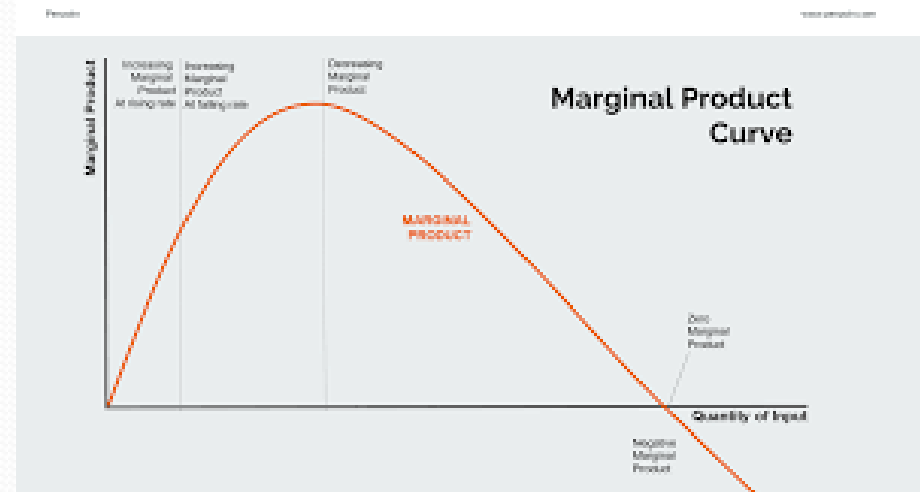
Total Product, Marginal product and Average product

- Total product (TP) is the total output produced with a given amount of factor inputs.
- The total product curve is S-shaped.
- the TP curve shows how total output varies with the quantity of labour, given the quantity of capital.



Marginal Product

- Marginal product (MP) is the additional output resulting from employing one more unit of labour.
- Marginal product is calculated by dividing the change in total output by the change in the quantity of labour.
- The marginal product curve is inverted-U-shaped.



Average product curve

- Average product (AP) is the output per unit of labour.
- Average product is calculated by dividing total output by the quantity of labour.
- The average product curve is inverted-U-shaped.

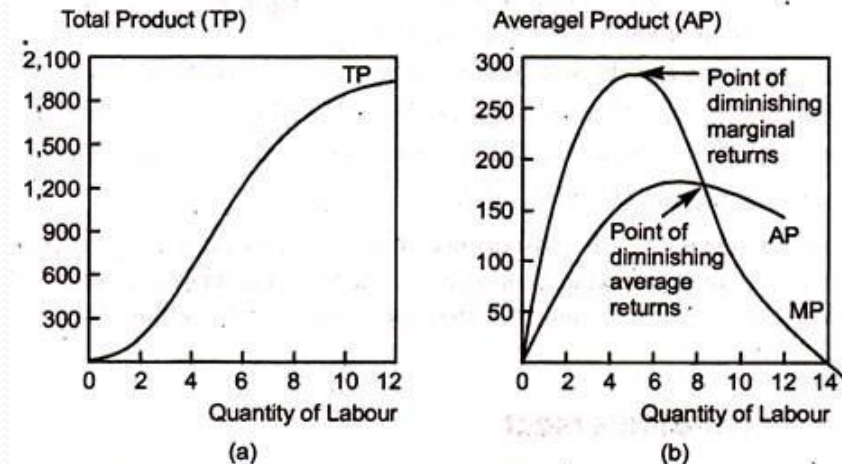


Fig. 6.2 : Relationship between Marginal and Average Product Curves

Explanation

- To understand the shape of the average product curve, we need to understand the relationship between average value and marginal value which can be illustrated with the following example.
- Assume that the average height of a particular class of students is 1.7 metres. Further assume that a new student joins the class. If the height of the new student, which is the marginal height, is higher than the average height of 1.7 metres, the average height of the class will rise. However, if the height of the new student is lower than the average height, the average height of the class will fall. Therefore, when the marginal value is higher than the average value, the average will rise and vice versa. This applies to the relationship between average product and marginal product

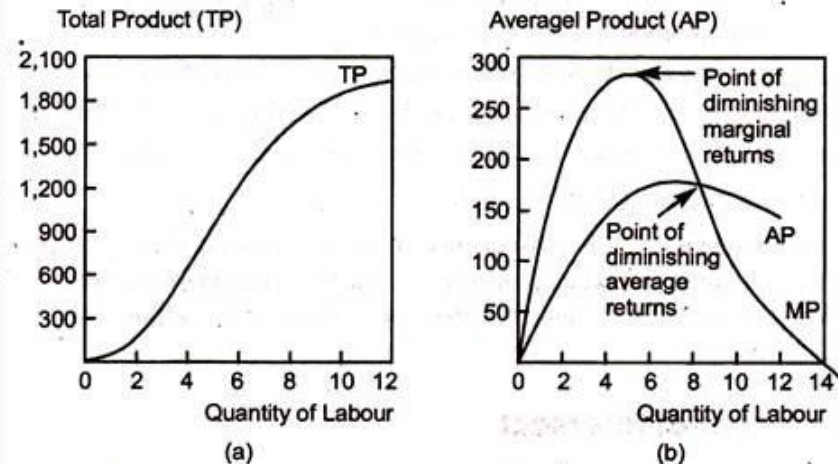


Fig. 6.2 : Relationship between Marginal and Average Product Curves

The Relationship between Average Product and Marginal Product

- When there is a rise in the average product due to an increase in the quantity of the variable input, the marginal product is more than the average product.
- The maximum average product is equal to the marginal product. Simply put, the maximum point of the average product curve is also a point on the marginal product curve, a point where both of these curves intersect.
- When the average product falls, the marginal product is less than the average product.

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- Now, the total productivity, for a given state of technology, is bound to increase with an increase in the quantity of a variable input. However, as the quantity of the inputs keeps on increasing, the marginal product rises to a maximum, then starts to decline and eventually becomes negative.

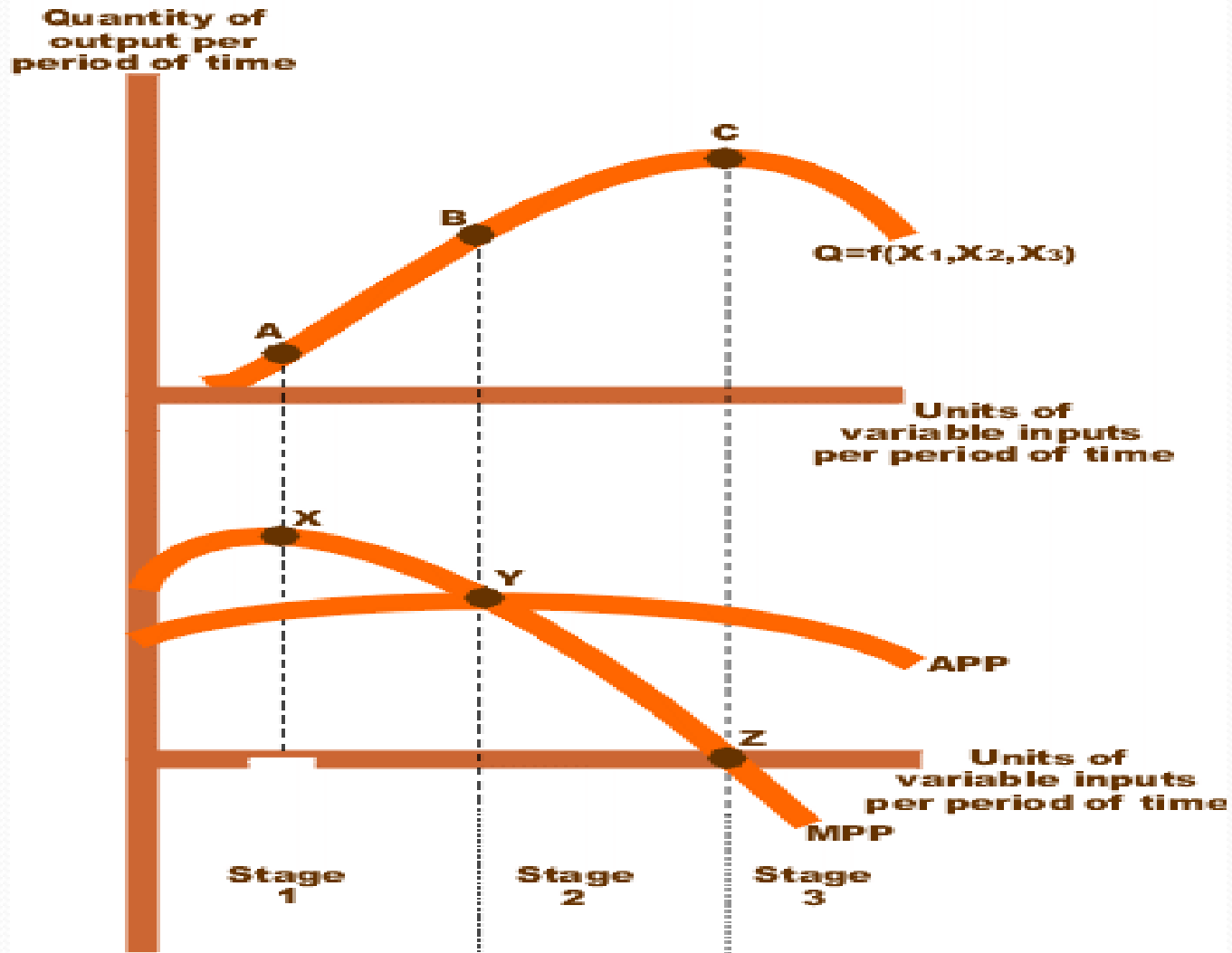
The law of diminishing returns also comes with some assumptions:

- We assume the state of technology to be constant. A variable state of technology would impact the marginal and average product. In that case, we would not be able to accurately study the relationship between output and the fixed input.
- Only one input should be variable. keeping other inputs constant. This law does not apply to cases when all the inputs vary proportionately. In that case, the returns to scale comes to the rescue.
- The law does not apply to a production scenario where we require specifically fixed proportions of inputs. In such a case, an increase in any input would not have any impact on production, since the marginal product will be equal to zero.
- We consider only physical inputs and outputs and not economic profitability in monetary terms.

The table for Law of diminishing returns

No. of Workers	Total Product (TP)	Average Product (AP)	Marginal Product (MP)
1	10	10	10
2	22	11	12
3	36	12	14
4	52	13	16
5	66	13.2	14
6	76	12.7	10
7	82	11.7	6
8	85	10.5	3
9	85	9.05	0
10	83	8.3	(-2)

Stages of production



Stage I: Increasing Returns

- We characterize this stage with the total output increasing at an increasing rate with each additional unit of the variable input. This continues to point A on the TP curve. Further, the MP curve rises to the point X corresponding to the point B on the TP curve, also known as the point of inflexion.
- After point B, the TP curve continues to rise but now at a decreasing rate. The MP also starts to fall but is positive. The end of this stage sees the maximum point of the average product, where the AP and MP curves intersect.

Stage I explained

- We get increasing returns in the first stage because initially, the fixed factors are abundant relative to the variable factor. The introduction of additional units of the variable factor leads to the effective utilisation of the fixed factors. Evidently, production increases at an increasing rate.

Stage II: Diminishing Returns

- Throughout the stage of diminishing returns, the total product keeps on increasing. However unlike the stage of increasing returns, here the total product increases at a diminishing rate. This happens because the marginal product falls and becomes less than the average product, which also sees a downwards slope.
- Thus, this stage is known as the stage of diminishing returns. The end of this stage is marked by the total product attaining its maximum value and the marginal product becoming zero. Further, this stage is very important because the firm will seek to produce in its range.

Stage III: Negative Returns

- The origin of stage 3 starts from the maximum point of the TP curve. In this stage, the TP curve now starts to decline. Moreover, the MP curve becomes negative coupled with a fall in the AP curve.
- The excessive addition of variable inputs leads to negative returns at this stage. This is because of the crowding of the variable factors. The variable and fixed factors now start getting into each other's ways. Effectively, there is no coordination and hence the output falls.

Stage of Operation

- A major dilemma in the world of the law of diminishing returns is deciding the stage where a rational producer would look to operate.

- Exercise:

At what stage a rational producer will operate?

Exercise:

- What is the behaviour of TP, MP, and AP at stage III?



